

Vol 5 Chapter 9 — Identical Particles and the Spin-Statistics Theorem

Keeper (author pass — deep math/physics revision)

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Chapter 9 — Identical Particles and the Spin-Statistics Theorem

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

Bosons and fermions are not independent particle classes but the two parity-sectors of the substrate's $Pin(2)$ double-cover structure: integer-spin K-types under particle exchange pick up $+1$ (bosons, symmetric); half-integer-spin K-types pick up -1 (fermions, antisymmetric) — the spin-statistics theorem is the substrate's $Pin(2)$ orientation rule applied to many-particle states.

Level 2 — graduate-physicist precision

9.1 Identical particles in standard QM

Two identical particles (e.g., two electrons): their wavefunction $\psi(x_1, x_2)$ must transform with definite sign under particle exchange $P_{12} : \psi(x_1, x_2) \mapsto \psi(x_2, x_1)$:

- Bosons: $P_{12}\psi = +\psi$ (symmetric)
- Fermions: $P_{12}\psi = -\psi$ (antisymmetric)

The Pauli exclusion principle for fermions follows: two fermions in the same single-particle state ϕ would give $\psi(x_1, x_2) = \phi(x_1)\phi(x_2)$, which is symmetric — incompatible with antisymmetry. So two fermions cannot occupy the same state.

9.2 The spin-statistics theorem

In relativistic QFT (Pauli 1940), the spin-statistics theorem establishes:

- Particles with integer spin ($s = 0, 1, 2, \dots$) are bosons
- Particles with half-integer spin ($s = 1/2, 3/2, \dots$) are fermions

The theorem requires Lorentz invariance, causality (microcausality), and positive-energy spectrum. The proof relies on the analytic continuation of field commutators / anticommutators between spacelike-separated regions.

In nonrelativistic QM, spin-statistics is an additional postulate (no derivation possible from Galilei invariance alone). In QFT, it's a theorem.

9.3 Substrate derivation via $Pin(2)$

BST's substrate provides a direct derivation through the $Pin(2)$ double-cover structure (Chapter 3 Section 3.5).

Recall: $Pin(2)$ is the double cover of $O(2)$; its representations are labeled by weights $k \in \frac{1}{2}\mathbb{Z}$. Half-integer k corresponds to “spinor” representations (electron, proton); integer k to “tensor” representations (photon, gluon).

For a substrate $Pin(2)$ representation with weight k , the 2π rotation generator $R(2\pi)$ acts as:

$$R(2\pi)|\text{spin } k\rangle = e^{2\pi i k}|\text{spin } k\rangle = (-1)^{2k}|\text{spin } k\rangle$$

So: - Integer k : $(-1)^{2k} = +1$ — unchanged under 2π - Half-integer k : $(-1)^{2k} = -1$ — picks up minus sign under 2π

This is the famous “spinor minus sign” — a 4π (not 2π) rotation returns a spinor to itself.

9.4 Particle exchange = π rotation in 2-particle space

For two identical particles, the exchange operator P_{12} can be realized as a continuous rotation in a fictitious 2-particle “exchange space” by angle π . The substrate $Pin(2)$ structure inherits this:

$$P_{12}|\text{two-particle, spin } k\rangle = e^{i\pi \cdot 2k}|\dots\rangle = (-1)^{2k}|\dots\rangle$$

Therefore: - Integer-spin particles: $P_{12}\psi = +\psi$ (bosons) - Half-integer-spin particles: $P_{12}\psi = -\psi$ (fermions)

The spin-statistics theorem follows from the substrate's $Pin(2)$ double-cover structure, without invoking relativistic QFT.

This is part of SP-31-15 (BST task #285): per-conservation-law substrate-derivation theorems include the spin-statistics theorem.

9.5 Many-particle states: Slater determinants and permanents

For N identical fermions in single-particle states $\phi_1, \dots, \phi_N$:

$$\Psi_{\text{fermion}}(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \phi_1(x_1) & \cdots & \phi_N(x_1) \\ \vdots & & \vdots \\ \phi_1(x_N) & \cdots & \phi_N(x_N) \end{pmatrix}$$

(Slater determinant). For bosons, replace determinant with permanent.

Substrate-mechanism reading: the substrate's K-type basis for N identical particles is the antisymmetrized (for fermions) or symmetrized (for bosons) tensor product of single-particle K-types. The Slater determinant is the substrate's natural antisymmetrization operation under $\text{Pin}(2)$ weight $1/2$.

9.6 Pauli exclusion and the periodic table

For N electrons in an atom: no two can occupy the same single-particle state (Pauli exclusion). This determines the electronic shell structure and the periodic table's row capacities $\{2, 8, 18, 32\}$ (Chapter 3 Section 3.7).

Substrate-mechanism: each substrate K-type (ℓ, m, s) accepts exactly one electron (one $\text{Pin}(2)$ half-integer-weight state). Row n has capacity $2n^2$ — twice the orbital-degeneracy sum, with factor 2 from spin doubling. The factor $2 = 2s + 1$ for $s = 1/2$ — the substrate's $\text{Pin}(2)$ minimum.

9.7 Statistics in atomic, nuclear, and exotic systems

- Atomic: electrons are fermions; periodic table follows.
- Nuclear: nucleons (protons, neutrons) are fermions; nuclear shell model (Mayer-Jensen 1949, L1 ESTABLISHED) has analogous shell structure. Volume 3 develops this.
- Bosonic: photons, mesons (e.g., pions), atomic gases below BEC temperature — symmetric statistics. Photon bunching, BEC.
- Anyons (2D systems): in 2D, exchange statistics can be neither symmetric nor antisymmetric but a complex phase (fractional statistics). Substrate-mechanism: in restricted 2D substrate sectors, the substrate's $\text{Pin}(2)$ double cover effectively becomes a $\pi_1(\text{exchange space})$ representation supporting fractional statistics. Connection to fractional quantum Hall effect (Volume 9 Chapter 10 Section 10.3).

9.8 Worked example: helium ground state

Helium has two electrons. The Pauli principle requires the total electron wavefunction to be antisymmetric:

$$\Psi(1, 2) = \psi_{\text{spatial}}(r_1, r_2) \cdot \chi_{\text{spin}}(1, 2)$$

with ψ_{spatial} symmetric $\times$ χ_{spin} antisymmetric (singlet) OR ψ_{spatial} antisymmetric $\times$ χ_{spin} symmetric (triplet).

Ground state: both electrons in 1s orbital, spatial part symmetric ($\phi_{1s}(r_1)\phi_{1s}(r_2)$), spin part antisymmetric singlet ($\chi_S = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$). Total: antisymmetric, as required.

Ground-state energy: $E_0 \approx -79$ eV (after electron-electron repulsion). Standard helium calculation; the substrate-mechanism reading is identical because Pauli exclusion = substrate Pin(2) antisymmetrization.

9.9 K-audit anchors

- T2471 / Casey-named principle #8 (Lyra Friday May 22, 2026): chirality from substrate Pin(2)
- SP-31-15 (BST task #285): spin-statistics theorem from substrate
- Mayer-Jensen 1949 (L1 ESTABLISHED): nuclear shell model, fermion statistics for nucleons
- Periodic table reading (Volume 0 Chapter 1): Pauli exclusion + Pin(2) spin doubling

Level 3 — 5th-grader accessibility

Particles come in two types: bosons (like photons) and fermions (like electrons). Bosons love to share — you can pile lots of them into the same state. Fermions hate to share — no two fermions can ever be in the exact same state. That’s why electrons stack up into shells in an atom: once a shell is full, the next electron must go to the next shell. The trick: bosons have integer spin (0, 1, 2, ...); fermions have half-integer spin (1/2, 3/2, ...). Why? Because of the substrate’s Pin(2) double cover: spinning a fermion by 360° multiplies its wave function by -1 , but spinning a boson by 360° multiplies by $+1$. Swap two fermions, you also pick up -1 . The periodic table’s rows (2, 8, 18, 32 electrons) are Pin(2) saying “fermions don’t share.”

What comes next

Chapter 10 develops decoherence and the classical limit — how the substrate’s Zone 3 commitment plus environment coupling produces classical macroscopic behavior.

Where to look this up

- Spin-statistics theorem: Pauli 1940; Streater and Wightman, *PCT, Spin and Statistics, and All That*
- Bose-Einstein statistics: Bose 1924, Einstein 1924-25
- Slater determinant: Slater 1929
- Anyons / fractional statistics: Wilczek 1990, *Fractional Statistics and Anyon Superconductivity*
- BST anchors: T2471 chirality, SP-31-15, Mayer-Jensen 1949 L1
- Volume 3 (Nuclear): nuclear shell model
- Volume 9 Chapter 10: fractional quantum Hall and anyons